Management of Change: Elekeiroz T-2103 butyraldehyde tower: recover the intended 20% capacity increase

DRAFT - NOT APPROVED. Prepared 2026-11-02 22:16 +03 by PlantAgent from the working model. Nothing in this file is an approval.

1. Summary

Decision requested	**Simulator screening passed; proceed to data and vendor verification (not ready for approval): 1 change to reach 121 % of today's throughput - (1) Replace T-2103 internals with high-capacity trays in same shell.**
Target	121 % of today's throughput (mixed butyraldehyde feed 9,500 → 11,500 kg/h)
Result (DWSIM, re-solved for this file)	At 121 %, existing plant: 1 limit over 100 % (highest T-2103 distillation Fair flood fraction vs limit 103 %); with the changes: all modelled process limits within 100 % (highest T-2103 outlet (product leaving the unit) temperature vs design 93 %).
Cost class (agent estimate, unpriced)	(1) $100k-$1M
Needs approval	(1) Replace T-2103 internals with high-capacity trays in same shell: cost class $100k-$1M (agent estimate), replacement rating (vendor confirmation open); Pre-startup safety review: 6 items (section 6.2).
Status	DRAFT for engineering review; to be checked by a licensed engineer. Sign-off in section 7.

Agent's explanation on record (model text, quoted; its figures are not re-checked here: sections 3 and 4 are the calculation record): 11.5 t/h needs high-capacity T-2103 trays. At 11.5 t/h the as-built 1.9 m T-2103 with 90 RV-1 trays floods at 102.6% (Fair, limit 0.80) in the calibrated run T2103_asbuilt_final_9p5_calib-2. With high-capacity internals rated at 1.30× RV-1 capacity the same shell passes at 78.9% and keeps isomer separation.

2. Basis

Question. T-2103 is limited near 9.5 t/h, while the project premise is 11.5 t/h. Build the existing tower, explain why added boil-up does not create useful capacity, and propose a design that reaches the premise without losing isomer separation.

Plant and data sources. Summers, Monaco and Britto, AIChE Spring Meeting paper 112b (2013), Sulzer/Elekeiroz; only the original tower and pre-solution operating evidence are exposed. https://www.sulzer.com/en/-/media/files/products/separation-technology/articles/butyraldehyde_tower_capacity_limitations.pdf

Data status. Real operating history and tower geometry. The paper does not publish a feed assay, so any binary i-/n-butyraldehyde composition used in DWSIM is an explicit modelling assumption. Later hardware and operating actions, plus the 2011-2012 performance data, are withheld in outcome.json.

Model. T2103_asbuilt_final_9p5_calib built by fireworks:accounts/fireworks/models/muse-spark-open on 2026-11-02 18:37; simulator DWSIM (headless). Review status: **ok** (0 findings, 0 blocking; details in appendix C).

3. Existing vs proposed at 121 % (DWSIM, re-solved for this file)

Item	Check	Limit	Existing	Proposed	Proposed value
T-2103 distillation	Fair flood fraction vs limit	0.8 fraction of flood	103 %	**79 %**	0.63 fraction of flood
T-2103 outlet (product leaving the unit)	temperature vs design	85 °C	93 %	**93 %**	79.4 °C
T-2103 outlet (product leaving the unit)	temperature vs design	80 °C	90 %	**90 %**	72.4 °C

Existing and Proposed are utilisation (value / limit) of each limiting item at the target; over 100 % fails. For an energy-closure check the limit is the heat available in that solve.

4. Changes

4.1 Replace T-2103 internals with high-capacity trays in same shell

Cloud 1 on the Rev B drawing. Cost class (agent estimate, unpriced): $100k-$1M.

What physically changes.

- Replaces T-2103 distillation: tray capacity factor 1.3 (High-capacity valve trays (e.g. MVG/Superfrac) rated 25-30% above RV-1 within existing 1900 mm dia, 425 mm spacing; retains 90-tray/64-ideal-stage basis per Summers et al 112b; vendor to confirm)

Why. Only T-2103 flooding limits the +21% (9.5→11.5 t/h). Same reboilers/condenser pass. DWSIM verification: flood 0.821→0.631, utilisation 102.6%→78.9% at 1.210526×.

Rating and rating source.

- T-2103: agent-proposed sizing basis (not independently validated): High-capacity valve trays (e.g. MVG/Superfrac) rated 25-30% above RV-1 within existing 1900 mm dia, 425 mm spacing; retains 90-tray/64-ideal-stage basis per Summers et al 112b; vendor confirmation open

Before / **Proposed calculation** (this change alone, DWSIM at the target).

- Simulated maximum: 118.1 % → 153.5 % of today; first limit T-2103 → T-2103.

Item	Check	Limit	Before	After
T-2103	Fair flood fraction vs limit	0.8 fraction of flood	0.82 fraction of flood (103 %)	0.63 fraction of flood (79 %)

Field checks that could change this.

- None raised for this change.

4.2 Other field checks

- Vendor hydraulic rating of HC trays vs RV-1 at 1.9 m, 425 mm spacing. Where: Tray vendor rating sheet for butyraldehyde system at 1.1-1.25 bar. If it differs: If CF <1.29, flood exceeds 80% and a larger CF or second tower is needed
- Actual reboiler duty margin at 11.5 t/h. Where: DCS steam to reboilers (6.9-7.3 t/h test range) vs design ~25% margin per paper. If it differs: Confirms reboilers not limiting; if margin smaller, add reboiler area

4.3 Economics and uncertainty of the design

Economics (ILLUSTRATIVE prices; order of magnitude)

- draft (assumed values): Capex ≈ $0k (class 5: $0k–$0k) plus unpriced items (replace T-2103): capex is a lower bound; product price not given: margin, payback and NPV open (energy and utilities $0k/yr). All prices ILLUSTRATIVE.

Proposal	Stated class	Factored capex	Check
Replace T-2103 internals with high-capacity trays in same shell	$100k-$1M	$0 + to price: replace T-2103	factored $0 is outside the stated class $100k-$1M

• Added Process feed: 48 t/d × 330 d; product margin $0/yr (solved 1.211 × base, capped at the 121% target; energy from dwsim solves (design at 121% vs today at 100 %)).
• Outage for tie-ins/cleaning: 2 d → $0 (muse_default shutdown for a tie-in or exchanger clean).
• Payback None years; NPV $0 (capex band: $0 to $0).

Uncertainty of the design (Monte Carlo over assumed and illustrative inputs)

• Target 121% met in 81% of 48 samples; 19% are over a limit at any throughput (T-2103: Fair flood fraction vs limit); max throughput P10/P50/P90 153% / 153% / 154% of the feasible samples; most often limited by T-2103: Fair flood fraction vs limit (81%); biggest driver limits.B_N.max_T_C (assumed).
• Method: Latin hypercube, 48 samples (seed 20261101), uniform ± band per input kind; each sample re-solved (dwsim): capacity bisection plus the solve at the target; 48 of 48 samples solved.

Input	Kind	Base	± band
units.T-2103.dP_bar	assumed	0.15	10 %
feeds.FEED.P_bar	assumed	2.0	10 %
feeds.FEED.T_C	assumed	65	10 %
limits.B_N.max_T_C	assumed	85	10 %
limits.D_ISO.max_T_C	assumed	80	10 %

• T-2103: Fair flood fraction vs limit at the target: load P10/P50/P90 79% / 79% / 79%.

5. Drawings

This project has no SVG P&ID; the changed equipment is: T-2103.

6. Risks and safety

6.1 Relief protection (P&ID relief scan)

- No smart P&ID: relief scan not available; review relief manually.

6.1c Discipline gates (outside the process model)

Pass/fail only from a sourced check; a check over an ILLUSTRATIVE rating leaves its gate open until the rating is confirmed. Electrical is in section 6.1b.

Gate	Status	Summary
robustness	**pass**	draft (assumed values): operating envelope 2/2 scenarios pass
piping utilities	**open**	no calculation adapter: tie-ins, line sizing, supports and utility headers remain open for T-2103
controls	**open**	no calculation adapter: control narrative, split/load-sharing control, cause & effect and permissives remain open for T-2103
process safety	**open**	no calculation adapter: change-focused HAZOP, relief case and isolation for the changed equipment remain open for T-2103
civil structural	**open**	no calculation adapter: plot, access, lifting and foundations remain open for T-2103

6.2 Pre-startup safety review (PSSR), generated from the changes

- [ ] Replacement of T-2103 distillation: confirm the installed rating (tray_capacity_factor = 1.3) against the vendor data sheet before start-up.
- [ ] Mechanical and process safety review for replacement T-2103: confirm materials, nozzles, supports, alarms, trips and relief basis are suitable for the new rating.
- [ ] Commissioning of replacement T-2103: update the equipment data sheet and procedures, then record the first operating point against the model.
- [ ] Update the P&ID, equipment datasheets and the operating procedure for every change above.
- [ ] Brief the operators on the new setpoints before start-up.
- [ ] Record the first 24 h of operation at the new throughput and compare with the model's predicted values in this file.

6.3 Second opinion

No second opinion was requested for this answer.

7. Approvals (DRAFT: not approved)

Role	Name	Signature	Date
Process engineer			
Operations			
Process safety (MOC)			
Manager (approver)			

Appendix A. Reproducibility

- Code version (git): 95ceeda; simulator: DWSIM (headless) via plantagent.process.
- Audit-chain head hash: `58bedf8b438432969a172ff369098f860b4ecec413f3e6c439c6af0f2abec864` (chain intact: True).
- Model: T2103_asbuilt_final_9p5_calib by fireworks:accounts/fireworks/models/muse-spark-open on 2026-11-02 18:37; answer by fireworks:accounts/fireworks/models/muse-spark-open.
- Run labels and calculation records: T2103_asbuilt_final_9p5_calib.
- Sections 1, 3 and 6.2 were recomputed when this file was made; re-run `python -m plantagent.decision revamp_butyraldehyde_t2103` on the same revisions to reproduce them.

Appendix B. Capability gaps

- None recorded.

Appendix C. Model review

Status: **ok**.

- No findings.

An overall energy balance is not closed: the simulator's streams do not report enthalpy. Unit duties are checked one by one instead.

Appendix D. Model assumptions

Recorded by the model builder when the model was built (model text). Later model corrections (section 2) take precedence; items the current model no longer matches are marked **superseded**.

- P&ID note: 'this project has no drawing: take the equipment and connections from the data sheet' - topology built from data_sheet text only; check_coverage returns 'no drawing to compare with' (as expected)
- Compound surrogate required: DWSIM compound database has no 'Isobutyraldehyde' / 'Isobutyraldehyde' key (KeyNotFound errors demonstrated). Butanal (n-butyraldehyde) + n-Hexane used as close-boiling isomer proxy with NRTL. Relative volatility tuned to ~1.15-1.20 range representative of iso-/n-butyraldehyde close split; NRTL selected per data_sheet instruction (UNIQUAC/NRTL for aldehydes). Labelled as assumption; absolute temperatures and duties are proxy-scaled - hydraulic vapour/liquid traffic and flood trend are the calibrated outputs.
- Feed composition and thermal condition not published (data_sheet: Feed composition was not published) - assumed equimolar proxy feed (x=0.5/0.5) at 65°C subcooled liquid, 2.0 bar feed pressure (pump discharge). Feed split, product purities and reflux ratio not specified; RR=6 / BR=3 chosen to give ~30% overhead split and hydraulic utilisation ~85% at 9.5 t/h, matching observed stable limit. All feed assay assumptions flagged as non-plant evidence per premise.
- Recycle topology: No external recirculation loop; tower is standalone with reboiler/condenser grouped per instruction '2 thermosyphon reboilers grouped'. Once-through feed->column->products modelled; no recycle unit needed (no tank mixing temperature set-point). Condenser duty computed, not limited.
- Tray rating: 90 actual trays *71% efficiency =63.9 ideal stages rounded to 64, feed stage 32 (mid-column). Spacing taken as midpoint 425 mm of 400-450 mm range; diameter 1,900 mm; flood limit 80% (Fair) with tray_spacing_m correction. DWSIM Fair correlation is sieve-tray based, not calibrated to RV-1 valve trays - utilisation is comparative only. Downcomer backup / vapour horn entrainment not expressible in steady-state stage model (see mechanism gap noted).
- Reboiler/condenser duties: Measured 4.1 M kcal/h (4768 kW) vs simulated ~2100 kW reflects surrogate latent heat / composition/purity difference, not a tuning failure; model preserves hydraulic vapour traffic trend (capacity_multiple 1.181) which matches intended 20% de-bottleneck to ~11.5 t/h. Design duty inferred as 4.1/0.75≈5.47 M kcal/h (6358 kW) not used as limit (no vendor rating provided).
- Pressure: Column P=1.1 bar top, dP=0.15 bar (1.25 bar bottom) assumed near-atmospheric operation typical for butyraldehyde; feed P=2.0 bar to ensure subcooled feed; product max_T/max_P limits set loosely (80-85°C) only to enable utilisation reporting.
- Utility not modelled: Steam flow 6.9-7.3 t/h cited only as heat proxy; no utility feed with utility:true and no max_m3_h limit to set (no supply limit given). Pressure drop rise <1% cannot be reproduced by fixed-dP stage model - reported as observed vs simulated gap, not tuned to match.